GEK-83844

DRIVE SYSTEM INSTRUCTIONS

CARE OF THE EQUIPMENT

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

SECTION - 2

CARE OF THE EQUIPMENT

(GEK - 83844)

HOW TO CARE FOR YOUR EQUIPMENT

RECEIVING, HANDLING, AND STORING

Receiving the Equipment

All equipment is carefully inspected and packed before leaving the factory: Upon receipt, an immediate inspection should be made to identify any missing or damaged items. If damage is evident or if there is visual indication of rough handling, a claim for damage should be filed immediately with the carrier. Both the transportation company and General Electric Company should be notified. Information as to damaged parts, part number, case number and requisition number should accompany the claim.

Handling the Equipment

Specific details, such as panel length, width, weight and lifting provisions, are included in the panel outline drawings. The center of gravity of the equipment should always be carefully observed to prevent tipping during handling.

Storing the Equipment

Equipment must be placed under adequate cover immediately. The packing coverings do not protect the equipment for outdoor storage.

The control equipment should be kept clean, dry and protected from extreme temperature variations, high humidity and dust. Variations in storage temperature can cause moisture condensation on the equipment. Maintaining the internal temperature of the equipment 5°C to 10°C higher than outside ambient will eliminate this condensation. This may best be accomplished by adding space heaters or using panel space heaters where supplied. This requires external AC supply voltage.

Equipment may be stored in environments of 5%-95% relative humidity with the above provisions to prevent condensation. Covering material should be breathable (canvas type); do not use plastic. Absolute storage temperature limits are between -40°C and 70°C.

OPERATING ENVIRONMENT

The following operating environment must be maintained to ensure proper operation and maximum equipment life:

- Operating temperature 0°C to 50°C
- Relative humidity 5% to 95% non condensing
- Altitude 3500 feet maximum without derating
- Corrosive gas content (parts per billion) maximum:

Hydrogen chloride	4 PPB
Chlorine	2 PPB
Hydrogen sulfide	2 PPB
Sulphur dioxide	4 PPB
Chlorine dioxide	4 PPB

- Dust

Particle size	10-100 microns
Concentration	0.0033 grams/cu.ft. air

CAUTION

THE EFFECT OF CORROSIVE GASES AND DUST IS ACCELERATED AT HIGH HUMIDITY AND, WITH CONDENSATION, CAN CAUSE ELECTRICAL SHORT CIRCUITS. FOR BEST RESULTS, KEEP EQUIPMENT COOL, DRY AND CLEAN.

PREVENTIVE MAINTENANCE

Section 8 of this instruction book contains publications for specific devices. The following comments are offered as general preventive maintenance recommendations.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. REMOVE ALL POWER FROM THE DRIVE SYSTEM BEFORE PERFORMING PREVENTIVE MAINTENANCE.

Inspection Routine

A periodic routine of equipment cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation due to the accumulation of dust and corrosion. Under normal conditions, the protective devices do not operate, therefore, these devices need careful inspection. If the equipment is in operation while being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will provide an opportunity to check its operation.

Log Book

It is recommended that a log book be kept of the regular inspections and maintenance. Specific items should be checked. Frequency of inspection will vary due, in a large part, to the fact that it is a function of environment and housekeeping practices.

MAINTENANCE RECOMMENDATIONS



ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

Loose Connections

Check for loose connections. Vibration may cause terminal board and bus bolt connections to become loose. Loose connections permit moisture and dust accumulations that can result in loss of low level signals at terminal board connections or thermal runaway at bus connections. Problems caused by loose connections may appear and disappear due to vibration.

Damaged Insulation

Check for damaged insulation on wires and cables. Insulation damage may cause intermittent shorting of signal or power cables.

Dust Accumulation

Check for dust accumulation on insulation and wire surfaces. Dust accumulating on stand-off insulators can collect enough moisture to produce a current path from bus bars to chassis ground. Dust (especially metallic dust) on wire surfaces can cause "tracking" between receptacle pins. Tracking is usually capacitive in nature and involves a build-up of electrical charge along the wire surface. This results in intermittent problems which are very difficult to locate. Clean bus bars and stand-off insulators with a clean dry cloth; Do not use any solvents. Clean wire surfaces between wiring receptacles by blowing dry air across all wire surfaces, directing the air source so that foreign objects are removed rather than relocated.

SCR Heat Sink Assemblies

Where very dirty conditions exist, there may be enough accumulation of foreign matter to restrict the air flow across the heat sink assemblies. Before performing any maintenance in the SCR power bridge, ensure that AC bridge power is locked out, To remove dirt from the heat sinks, clean them with a strong industrial vacuum cleaner using a brush attachment.

Printed Wiring Boards

Boards can be cleaned in a lukewarm (100°F) solution of mild household detergent or mild solution of a laboratory detergent like ALCONOX or NEUTRA-CLEAN. Use a soft-bristled brush such as a paint brush to scrub the board. After cleaning, rinse thoroughly in several baths of lukewarm, fresh water. Shake off excess water, immerse the board in isopropyl alcohol and agitate, then air dry for a few hours. Do not use solvents containing ammonia, aldehydes, alkalies, aromatic hydrocarbons, or ketones.

Short Circuits

Normally, the overcurrent protective device on the circuit will prevent any electrical damage except at the point of a short-circuit. After any large fault current be sure to make a thorough inspection of the entire system to ensure that there has been no mechanical damage to conductors, insulation or equipment. In addition, the individual overcurrent protective device or devices which performed the short-circuit interruption must be inspected for possible arcing damage to contacts, arc chutes and/or insulation.

Arcing Damage To Insulation

Some organic insulating materials carbonize and lose their insulating qualities when subjected to the heat of an electrical arc. Any insulation found to be carbon-tracked must be replaced before applying power.

Clean Filters

Power Converter door mounted filters should be cleaned regularly to ensure constant air flow.

DC Motors

See motor manufacturer's manuals for maintenance instructions.

NOTES

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